



DEPARTMENT OF APPLIED COMPUTING & ARTIFICIAL INTELLIGENCE

Faculty of Computing

SECP3106 APPLICATION DEVELOPMENT (WBL)

Section 01

INDUSTRY PROJECT PROPOSAL

in collaboration with

TVETMARA Johor Bahru (Institut Kemahiran Mara Johor Bahru)

Group CodeCraft

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Submission Date: 28th April 2026

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1.0 Project Background

TVET MARA Johor Bahru (MARAJB) manages a significant academic population of 800 students overseen by 110 academic staff. To improve administrative efficiency, a centralized Internal Web-Based Learning Management System (LMS) is required. Unlike traditional student-facing portals, this system is designed exclusively as a tool for staff and administrators to manage student records, attendance, and academic progression more effectively.

2.0 Objectives

The objective of this project is to develop a centralized and automated academic management system that improves efficiency by integrating attendance tracking, semester management, evaluations, examinations, and part-time course processing into a single platform. The system aims to reduce manual work, enhance data accuracy, and streamline administrative workflows.

3.0 Target User

The proposed system is designed exclusively for the internal staff of TVET MARA Johor Bahru. While students are the subjects of the data, they do not have login access. The users are categorized into the following three primary roles:

3.1 Academic Lecturers

The lecturers are the primary end-users responsible for the daily input of academic data. Their use of the system includes:

- **Attendance Tracking:** Recording daily student attendance and monitoring the 80% threshold for exam eligibility.
- **Grade Management:** Inputting marks for continuous assessments (30%) and final examinations (70%), including theory, practical (amali), and oral components.
- **Student Monitoring:** Updating individual student statuses and identifying those requiring academic intervention.

3.2 Heads of Programs (*Ketua Program*)

The Heads of Programs serve as the administrative authority within the system. Their roles include:

- **Approval Authority:** Reviewing and approving student promotions to the next semester or year.
- **Status Validation:** Formally authorizing changes in student status, such as deferrals (*Tangguh*), dismissals (*Dibuang*), or withdrawals (*Berhenti*).
- **Quality Oversight:** Monitoring the lecturer evaluation (eppp) results and average marks for their respective departments.

3.3 Administrative & Part-Time Course Coordinators

This group focuses on the specialized modules for industry-linked training and general administration:

- **Course Coordination:** Managing registrations and records for part-time industry courses (e.g., AutoCAD examination).
- **Document Generation:** Utilizing the system to automate the creation of offer letters and student registration slips.
- **Legacy Data Management:** Accessing and migrating historical data from previous MDB formats into the new web-based environment.

4.0 Problem Statement

The current education management and learning management processes at TVETMARA Johor Bahru are highly decentralised. They are relying heavily on manual data input and outdated systems which leads to serious operational inefficiency. The important academic data is stored in a decentralised manner, and the exam results are isolated and stored in outdated Microsoft

Access database files which need to be read manually, while the teaching and learning process (T&L) is tracked separately in an independent OneNote document.

In addition, manual forms are required to record students' attendance which may be a time-consuming process. Senior management tasks do not automate workflows such as the head of department's approval of student identity changes such as “deferred” or “suspended”, and the screening of applicants for part-time industry courses. Besides, there is no mandatory system mechanism to ensure that students complete the required instructor evaluation before obtaining the examination slip.

5.0 Proposed Solution

In order to solve these inefficiency problems, our team proposes to develop a centralised application presented entirely in *Bahasa Melayu* to integrate these academic and administrative workflows so to reduce staff retraining. The system will adopt an automated data pipeline design to read and automatically fill in students' grades directly from existing Microsoft Access files and extract teaching progress tracking information from OneNote.

The core operation will be modernised by providing a digital attendance interface that matches the manual form. We will adopt a simple and clear framework of “1 as present”, “0 as absent” and “leave with approval” to achieve a seamless transition. In addition, we will implement a secure automatic approval process to manage students' promotion of semester matters and set up an automatic filtering function to verify part-time course applicants according to the minimum requirements to automatically generate official admission notices.

To ensure compliance with the policy, this application will also incorporate conditional logic, which means that the permission to download the examination slip will be automatically blocked until the student successfully submits the lecturer evaluation.

6.0 Mandatory Module

6.1 Login System

The login system provides users with authentication methods to access the application. As the main security entry point, it will verify user credentials before granting access to the system. The system will also support role-based routing to ensure that students, instructors and administrators are automatically guided to their respective dashboards after successful authentication.

6.2 Registration System

The registration system facilitates the registration process for new users. This module will perform basic data verification to ensure that the format of the data entered such as student ID, employee number and password is correct before creating a new account in the registration database.

6.3 Manage User System

This User Management system is the central management centre for account management. This module enables authorised administrators to securely view, edit, or delete user accounts across the platform. In order to maintain strict data security and operational integrity when managing employee and student lists, the system will implement role-based access control. This ensures that management capabilities are appropriately limited based on user levels, such as students, instructors, and administrators, and only grant them permissions that match their specific roles.

7.0 Industry Module

7.1 *Sistem Kedatangan Pelajar*

This module is the first of our six core industry-specific modules. This is a digital attendance tracking tool specially designed for lecturers. The user interface will adopt a simple checkbox system to ensure a fast and intuitive user experience. When the lecturer checks the checkbox

to indicate attendance, the backend of the system will automatically mark “1” as present and “0” as absent for no reason. For authorised and approved leave, the system still marks the value as “1”, but will add specific notes to distinguish it from regular attendance. In addition, this module will also include an automatic reporting function, which can calculate and display the total attendance rate of each student at the end of the semester.

7.2 *Sistem Penukaran Semester*

This module manages student progression between semesters with bulk review and approval. It assigns outcomes such as continue to next semester (*teruskan pengajian*), defer (*ditangguh*), stop (*berhenti*), or expel (*dibuang*), subject to Head of Program approval.

7.3 *Sistem Penilaian Pensyarah*

This module manages lecturer evaluations for each subject, where evaluations must be completed before the final examination slip can be accessed. It also allows lecturers to view the average evaluation percentage of each subject they taught, which helps to better comprehend students’ feedback and constantly enhance the quality of teaching.

7.4 *Sistem Penilaian Pembelajaran dan Pengajaran*

This module manages the automated collection and analysis of lecturer performance data by linking student evaluations (EPPP) directly to the issuance of official academic documents. It functions as a mandatory administrative checkpoint where the system restricts staff from generating course registration slips until a student’s evaluation record is confirmed as complete. By centralizing this workflow, the module eliminates manual calculations by automatically aggregating student feedback into average performance marks, allowing the management to monitor teaching effectiveness and ensure 100% compliance before students proceed to the next semester’s registration.

7.5 *Sistem Peperiksaan ST*

This module was created to oversee student test procedures in a more effective and organized manner. After meeting prerequisites, including completing instructor evaluations, students can access and download their examination slip online. Administrators can also manage student eligibility, upload exam schedules, and automatically create test slips with this module. Without the need for human verification, students can access their exam details via the system, including the date, time, location, and registered subjects. Furthermore, the solution reduces reliance on out-of-date Microsoft Access files and minimizes human error in managing examination data by securely storing examination records in a centralized database.

7.6 *Sistem Kursus Separuh Masa*

This module manages applications for part-time courses (e.g., AutoCAD) by checking eligibility criteria such as industry experience, computer literacy, and Malaysian status. Eligible applicants are automatically processed and issued an official offer letter.

8.0 **Module Distribution**

Module	Type	Lead Member
Login	Mandatory	Lau Yan Kai
Registration	Mandatory	Lau Yan Kai
Manage User	Mandatory	Lau Yan Kai
Student Attendance (<i>Kedatangan Pelajar</i>)	Industry	Lau Yan Kai
Semester Transfer (<i>Penukaran Semester</i>)	Industry	Joanne Ching Yin Xuan
Lecturer Evaluation (<i>Penilaian Pensyarah</i>)	Industry	Evelyn Goh Yuan Qi
Teaching and Learning Evaluation (<i>Penilaian Pembelajaran dan Pengajaran</i>)	Industry	Syarifah Dania binti Syed Abu Bakar
ST Examination (<i>Peperiksaan ST</i>)	Industry	Damiya Aina binti Basir Abd Shammad
Part-Time Course (<i>Kursus Separuh Masa</i>)	Industry	Joanne Ching Yin Xuan

9.0 Justification

The existing TVETMARA Johor Bahru management procedures heavily rely on human labor, antiquated software, and decentralized data storage, which justifies the proposed centralized academic management system. These problems frequently result in delays, redundant work, human error, and challenges with effectively managing crucial academic records. All academic and administrative modules, including attendance monitoring, semester management, lecturer evaluations, exam administration, and part-time course applications, may be managed on a single platform thanks to the development of this integrated system. Staff workloads are lessened, operational effectiveness is increased, and information may be accessed more quickly.

Additionally, automation capabilities like digital record management, eligibility checks, and approval workflows can reduce paperwork while increasing accuracy. A more user-friendly system that makes services and information easier to access at any time would also help instructors and students. Additionally, by making Bahasa Melayu the primary language of the application, TVETMARA Johor Bahru customers are guaranteed familiarity and convenience of use, which lessens the need for retraining. All things considered, this initiative is acceptable because it promotes digital transformation, boosts efficiency, strengthens data security, and offers a cutting-edge replacement for antiquated, ineffective procedures.

10.0 Conclusion

In conclusion, our team aims to successfully develop a functional and integrated academic management application that includes all mandatory and industry modules by the end of the semester. The system is anticipated to automate academic and administrative processes at TVETMARA Johor Bahru, enhance data integrity and efficiency of the system, and improve the overall user-friendliness of the system.